

Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data without being explicitly programmed.

Instead of writing rules manually, ML algorithms identify patterns from historical data and use those patterns to make predictions.

Types of Machine Learning

Supervised Learning

Uses labeled data.

Examples:

- House price prediction
- Email spam detection

Unsupervised Learning

Uses unlabeled data.

Examples:

- Customer segmentation
- Clustering

Reinforcement Learning

The model learns by receiving rewards and penalties.

Examples:

- Robotics
- Game Playing

Applications

- Fraud Detection
- Recommendation Systems
- Predictive Analytics
- Medical Diagnosis